Deep Learning Lab: Viva Questions

Experiments 1 to 11 (TensorFlow). 20 questions with short answers for each experiment.

Experiment 1: Image Processing (OpenCV)

Q1. What is histogram equalization?
It spreads pixel intensities over the full 0 to 255 range to improve contrast.

Q2. Why does it help dark or low-contrast images?
Their intensities are bunched in a narrow band. Equalization stretches them, so details become visible.

Q3. What does an image histogram show?
The number of pixels at each intensity value (0 to 255).

Q4. Which function equalizes an image?
cv2.equalizeHist(). It works only on an 8-bit single-channel (grayscale) image.

Q5. Why is the image read in grayscale?
Equalization, Canny and thresholding work on one channel. It also means less data.

Q6. What is Gaussian blur?
Smoothing with a Gaussian kernel. It reduces noise.

Q7. Why blur before Canny?
Noise creates false edges. Blur removes the noise first.

Q8. What is Canny edge detection?
A multi-stage method: gradient calculation, non-maximum suppression, double threshold, edge tracking by hysteresis.

Q9. What do 100 and 200 mean in Canny?
Low and high thresholds. Above 200 is a strong edge. Below 100 is dropped. In between is kept only if connected to a strong edge.

Q10. What is thresholding?
It converts a gray image to binary. Pixels above the threshold become 255, others become 0.

Q11. What does cv2.threshold return?
Two values: the threshold used and the binary image.

Q12. What is erosion?
It shrinks white regions and removes small white noise.

Q13. What is dilation?
It grows white regions and fills small holes.

Q14. What is the role of the kernel np.ones((5,5))?
It is the structuring element. Its size decides how much the shape shrinks or grows.

Q15. Opening vs closing?
Opening = erosion then dilation (removes noise). Closing = dilation then erosion (fills holes).

Q16. What is data augmentation?
Creating changed copies of training images to enlarge the dataset.

Q17. Why do we augment data?
It reduces overfitting and improves generalization.

Q18. Which class does the augmentation here?
ImageDataGenerator with rotation_range=40, zoom_range=0.2, horizontal_flip=True.

Q19. Why convert BGR to RGB?
cv2.imread gives BGR but matplotlib expects RGB. Use cv2.cvtColor(img, cv2.COLOR_BGR2RGB).

Q20. Why astype('uint8') on augmented images?
The generator returns floats. imshow needs integers from 0 to 255 to show correct colours.

Experiment 2: ANN on MNIST

Q1. What is an ANN?
Layers of neurons with weights that learn a mapping from input to output.

Q2. What is the MNIST dataset?
70,000 grayscale 28 x 28 digit images: 60,000 for training, 10,000 for testing, 10 classes.

Q3. Why divide pixels by 255?
It scales values to 0 to 1, so training is faster and more stable.

Q4. Purpose of Flatten?
It turns the 28 x 28 image into a 784 vector for the Dense layer.

Q5. What is ReLU?
max(0, x). It adds non-linearity and avoids vanishing gradients.

Q6. Why softmax in the output layer?
It converts 10 scores into probabilities that sum to 1.

Q7. Why 10 neurons in the output layer?
One for each digit class (0 to 9).

Q8. Parameters in Dense(64) after Flatten?
784 x 64 + 64 = 50,240.

Q9. Parameters in the output Dense(10)?
64 x 10 + 10 = 650.

Q10. What is the Adam optimizer?
An optimizer with adaptive learning rates that uses momentum and the average of squared gradients.

Q11. Why sparse categorical crossentropy?
Labels are plain integers (0 to 9), not one-hot vectors.

Q12. How is it different from categorical crossentropy?
Categorical crossentropy needs one-hot labels.

Q13. Epoch vs batch size?
Epoch = one full pass over the data. Batch size = samples per weight update (32 here).

Q14. What does validation_split=0.2 do?
It holds out 20% of the training data to check performance after every epoch.

Q15. Training vs validation vs test data?
Training fits the weights. Validation monitors and tunes. Test gives the final unbiased score.

Q16. How do curves show overfitting?
Training accuracy keeps rising while validation accuracy stalls or drops. The gap grows.

Q17. What does model.evaluate return?
The loss and the accuracy on the test data.

Q18. What is backpropagation?
It finds the gradient of the loss for each weight using the chain rule. The optimizer then updates the weights.

Q19. Why set a random seed?
To get the same results on every run.

Q20. How can accuracy be improved?
More layers or neurons, Dropout, more epochs, or use a CNN.

Experiment 3: CNN on EMNIST Balanced

Q1. What is a CNN?
A network that uses convolution filters to learn spatial features from images.

Q2. What is EMNIST Balanced?
Handwritten letters and digits with 47 classes and equal samples per class (112,800 train, 18,800 test).

Q3. Why 47 classes and not 62?
Similar upper and lower case letters (like c, o, s, v, w, x, z) are merged: 10 digits + 26 + 11 = 47.

Q4. What does Conv2D(32, 3) mean?
32 filters, each of size 3 x 3.

Q5. Output size of the first Conv2D on 28 x 28?
26 x 26, because there is no padding (28 - 3 + 1).

Q6. What is MaxPooling?
It takes the maximum in each 2 x 2 window. It halves the size, cuts computation and gives small shift tolerance.

Q7. Why two Conv + Pool blocks?
The first learns simple edges. The second combines them into shapes.

Q8. Purpose of Flatten?
It converts the feature maps to a 1D vector for the Dense layers.

Q9. What does Dropout(0.5) do?
It randomly switches off 50% of neurons during training to reduce overfitting. It is off at test time.

Q10. Why is a CNN better than an ANN for images?
It keeps spatial structure and shares weights, so it needs far fewer parameters.

Q11. Why input shape (28, 28, 1)?
Height, width and 1 channel (grayscale).

Q12. Why reshape to add a channel?
Conv2D needs 4D input: (batch, height, width, channels).

Q13. How is the .gz idx file read?
gzip.open then np.frombuffer. Skip a 16-byte header for images and an 8-byte header for labels.

Q14. What is in the idx header?
Magic number, item count and, for images, rows and columns.

Q15. What is the mapping file?
It maps a label number to the ASCII code of its character. chr() converts it to a letter.

Q16. What does model.predict return here?
47 probabilities. argmax gives the class and max gives the confidence.

Q17. Why softmax with 47 outputs?
It gives a probability for each of the 47 classes.

Q18. Parameters in the first Conv2D?
(3 x 3 x 1 + 1) x 32 = 320.

Q19. Why only 4 epochs?
The dataset is large, so the model learns fast and training time stays short.

Q20. Why do EMNIST images look rotated?
The original files store them transposed. The model still learns, since train and test have the same orientation.

Experiment 4: CNN Face Recognition (LFW)

Q1. What is the LFW dataset?
Labeled Faces in the Wild: face photos of public figures collected from the web.

Q2. What does min_faces_per_person=70 do?
It keeps only people with at least 70 images, so no class has too few samples.

Q3. What does resize=0.5 do?
It scales images to half size (62 x 47), so training is faster.

Q4. Why add a channel dimension?
Images are grayscale. Conv2D needs shape (62, 47, 1).

Q5. What does stratify=y do?
It keeps the class proportions the same in train and test sets.

Q6. Meaning of test_size=0.2 and random_state=42?
20% of data is for testing. random_state makes the split repeatable.

Q7. Face recognition vs face verification?
Recognition says who the person is. Verification says if two faces are the same person.

Q8. Why can accuracy mislead on this data?
The classes are imbalanced (one person has many more images). Use F1 score or a confusion matrix too.

Q9. Why is the same CNN as Exp 3 used?
The task is the same type (image classification). Only input shape and class count change.

Q10. How many output neurons are used?
len(names), one per person.

Q11. Why normalize by 255?
It keeps inputs in 0 to 1 for stable training.

Q12. What is a feature map?
The output of one filter applied over the image.

Q13. What does pooling do for faces?
It keeps the strongest features and reduces size.

Q14. What is overfitting and how to reduce it?
The model memorizes training data. Use Dropout, augmentation, more data or early stopping.

Q15. What is a confusion matrix?
A table of actual vs predicted classes.

Q16. How to predict a new face?
Resize to the same shape, divide by 255, add a batch dimension, call predict, then argmax gives the name.

Q17. Why argmax on the output?
It picks the class with the highest probability.

Q18. Classical alternative to a CNN?
Eigenfaces (PCA) with an SVM classifier.

Q19. What is transfer learning?
Reusing a pretrained network (like ResNet or VGGFace). It works better with little data.

Q20. Uses and concerns of face recognition?
Uses: phone unlock, security, attendance. Concerns: privacy and bias.

Experiment 5: Character-Level Text Generation (SimpleRNN)

Q1. What is an RNN?
A network with a loop. A hidden state carries information from earlier steps, so it suits sequences.

Q2. What is a SimpleRNN layer?
The basic RNN layer: h(t) = tanh(W x(t) + U h(t-1) + b).

Q3. What is a character-level model?
It predicts the next character from the previous characters.

Q4. Character-level vs word-level?
Character-level has a small vocabulary and no unknown words, but sequences are longer and meaning is weaker.

Q5. Why convert characters to integers?
Networks need numbers. We build char-to-index and index-to-char dictionaries.

Q6. What are the input and target?
Input is a fixed-length chunk of characters. Target is the next character.

Q7. Why use one-hot or an embedding?
To represent each character as a vector the network can use.

Q8. What is the output layer?
A Dense layer with softmax over the vocabulary size.

Q9. Which loss is used?
Categorical crossentropy (sparse version if labels are integers).

Q10. What is the vanishing gradient problem?
Gradients shrink over long sequences, so SimpleRNN forgets long-term context.

Q11. How do we fix it?
Use LSTM or GRU.

Q12. What is temperature in sampling?
It controls randomness. Low is safe and repetitive. High is creative but noisy.

Q13. Greedy vs sampling?
Greedy (argmax) always gives the same text. Sampling gives variety.

Q14. What is the seed text?
The starting characters used to begin generation.

Q15. How is text generated?
Predict the next character, append it, slide the window, repeat.

Q16. Why use tf.data.Dataset?
It builds a fast, batched and shuffled input pipeline.

Q17. What is backpropagation through time?
Backpropagation applied across the unrolled time steps of an RNN.

Q18. What is perplexity?
A measure of how well the model predicts text. Lower is better.

Q19. Why is early output nonsense?
Few epochs and a small model. It improves with more training.

Q20. Applications of text generation?
Autocomplete, chatbots, music and code generation.

Experiment 6: IMDB Sentiment Analysis (LSTM)

Q1. What is the IMDB dataset?
50,000 movie reviews (25,000 train, 25,000 test) labeled positive or negative.

Q2. What type of problem is it?
Binary classification. Usually one sigmoid output with binary crossentropy loss.

Q3. How are reviews stored in Keras?
As lists of integers. Each integer is a word index ranked by frequency.

Q4. Why limit the vocabulary (num_words)?
It keeps only the most frequent words. It saves memory and removes rare noise.

Q5. Why pad sequences?
Batches need equal length, so reviews are padded or cut to maxlen.

Q6. What is an Embedding layer?
It learns a dense vector for each word. Similar words get nearby vectors.

Q7. Why Embedding instead of one-hot?
It is much smaller and captures meaning.

Q8. What is an LSTM?
An RNN with gates and a cell state that remembers long-range information.

Q9. Name the three LSTM gates.
Forget gate, input gate and output gate.

Q10. Cell state vs hidden state?
Cell state is the long-term memory. Hidden state is the output at each step.

Q11. Why LSTM over SimpleRNN?
It reduces the vanishing gradient problem.

Q12. Why sigmoid and tanh in an LSTM?
Sigmoid (0 to 1) works as a gate. Tanh (-1 to 1) creates the values.

Q13. What threshold gives the class?
0.5. Above it is positive, below it is negative.

Q14. How to reduce overfitting here?
Dropout, fewer epochs or a smaller model.

Q15. How many parameters in the Embedding layer?
Vocabulary size x embedding dimension.

Q16. Formula for LSTM parameters?
4 x (units x (input_dim + units) + units).

Q17. What is a Bidirectional LSTM?
It reads the text forward and backward for better context.

Q18. Is accuracy a good metric here?
Yes, because the classes are balanced (equal positive and negative reviews).

Q19. How to predict a new review?
Convert words with imdb.get_word_index (offset of 3), pad it, then call predict.

Q20. Applications of sentiment analysis?
Product reviews, tweets, customer feedback and brand monitoring.

Experiment 7: POS Tagging (Encoder-Decoder LSTM)

Q1. What is POS tagging?
Labeling each word with its part of speech.

Q2. Give an example.
The cat sleeps -> DET NOUN VERB.

Q3. Why is POS tagging useful?
It helps parsing, named entity recognition, translation and search.

Q4. Sequence labeling vs classification?
Sequence labeling gives one label for every word, not one for the whole input.

Q5. What is an encoder-decoder model?
The encoder reads the input and makes a state. The decoder uses that state to produce the output.

Q6. What does the encoder pass on?
Its final hidden state and cell state (the context).

Q7. What does the decoder do?
It starts from the encoder states and predicts the tags.

Q8. Why pad sentences?
Sentences have different lengths. Padding makes them equal to the maximum length.

Q9. How are words and tags tokenized?
Each word and each tag gets an integer id using word-to-index and tag-to-index dictionaries.

Q10. Why an Embedding layer for words?
It turns word ids into dense vectors that carry meaning.

Q11. What is the output layer?
A softmax over the tag set at every time step.

Q12. Which loss is used?
Sparse categorical crossentropy, because tags are integers.

Q13. Why mask the padding?
So padded positions do not affect the loss or the accuracy.

Q14. How are unknown words handled?
They are mapped to a special UNK token.

Q15. What is the evaluation metric?
Token-level accuracy: correct tags divided by total tags.

Q16. Why is accuracy usually high?
Most words have a common tag. A most-frequent-tag baseline already gets about 90%.

Q17. What is teacher forcing?
Feeding the true previous output to the decoder during training.

Q18. How is it different from translation?
Here the output length equals the input length.

Q19. Other methods for POS tagging?
HMM, CRF, BiLSTM and Transformers (BERT).

Q20. How does the model handle ambiguity like 'book'?
The LSTM uses the surrounding words to decide if it is a noun or a verb.

Experiment 8: English to Tamil Translation (Seq2Seq LSTM)

Q1. What is Seq2Seq?
An encoder and a decoder that map an input sequence to an output sequence of a different length.

Q2. What is machine translation?
Automatic conversion of text from one language to another.

Q3. Why use start and end tokens?
The start token begins decoding. The end token tells the decoder to stop.

Q4. What is teacher forcing?
During training the decoder gets the true previous word instead of its own prediction.

Q5. Training vs inference decoder?
Training uses true words. Inference feeds back its own prediction, so separate encoder and decoder models are built.

Q6. What is the context vector bottleneck?
All input information is squeezed into one fixed vector. Long sentences lose information.

Q7. How is the bottleneck fixed?
With attention.

Q8. What is attention?
The decoder looks at all encoder outputs with learned weights at every step.

Q9. Why is Tamil harder than English?
It is agglutinative with rich word forms, has SOV word order and uses a different script.

Q10. Why two tokenizers?
English and Tamil have separate vocabularies.

Q11. Why is vocabulary size words + 1?
Index 0 is kept for padding.

Q12. Why an Embedding layer?
It gives dense vectors for words instead of sparse one-hot vectors.

Q13. What is the output layer?
A softmax over the Tamil vocabulary.

Q14. Which loss is used?
Sparse categorical crossentropy.

Q15. Why train for many epochs (100)?
The dataset is small, so the model needs many passes. Too many can overfit.

Q16. What is the BLEU score?
It measures n-gram overlap between the model output and a reference translation.

Q17. Greedy vs beam search?
Greedy takes the best word at each step. Beam search keeps several best paths and is usually better.

Q18. Why handle Unicode carefully?
Tamil text needs UTF-8 when reading and writing files.

Q19. Why pad both languages?
Encoder and decoder inputs need fixed lengths: the max English length and the max Tamil length.

Q20. What is the modern alternative?
The Transformer, which uses self-attention instead of recurrence.

Experiment 9: GAN Data Augmentation (Fashion-MNIST)

Q1. What is a GAN?
Two networks trained against each other: a generator and a discriminator.

Q2. What does the generator do?
It turns random noise into a fake image.

Q3. What does the discriminator do?
It classifies an image as real or fake.

Q4. What is the latent vector?
The random noise input to the generator, usually from a normal distribution.

Q5. What is Fashion-MNIST?
70,000 grayscale 28 x 28 clothing images in 10 classes.

Q6. What is the GAN objective?
A minimax game: the discriminator tries to win and the generator tries to fool it.

Q7. Which loss is used?
Binary crossentropy. Discriminator: real = 1, fake = 0. Generator wants the discriminator to say 1.

Q8. Why are the two networks trained alternately?
One is frozen while the other updates, so each learns against a fixed opponent.

Q9. What is mode collapse?
The generator makes only a few kinds of images and loses variety.

Q10. Why are GANs hard to train?
They are unstable. One network can overpower the other and gradients vanish.

Q11. Which output activation does the generator use?
Tanh if images are scaled to -1 to 1. Sigmoid if scaled to 0 to 1. Follow your code.

Q12. How does a GAN help data augmentation?
It creates new synthetic images to enlarge the training set.

Q13. Why GAN instead of flips and rotations?
It makes new varied samples and can help class imbalance.

Q14. How to check generated image quality?
Look at them, use FID, or test a classifier trained on the augmented data.

Q15. How to test if augmentation helped?
Train a classifier on real data and on real + fake data, then compare test accuracy.

Q16. What do Conv2DTranspose layers do?
They upsample feature maps. The generator uses them to build larger images.

Q17. Why LeakyReLU and BatchNorm in GANs?
LeakyReLU avoids dead neurons. BatchNorm stabilizes training.

Q18. Name some GAN types.
DCGAN, cGAN, WGAN and CycleGAN.

Q19. What is a conditional GAN?
A GAN conditioned on a class label to generate a chosen class.

Q20. Uses and risks of GANs?
Uses: augmentation, super-resolution, art. Risk: deepfakes.

Experiment 10: Recommendation System (Embeddings)

Q1. What is a recommender system?
A system that suggests items a user is likely to want.

Q2. Types of recommender systems?
Content-based, collaborative filtering and hybrid.

Q3. Which type is this model?
Neural collaborative filtering using embeddings.

Q4. What is an embedding?
A learned dense vector for each user or product.

Q5. Why embedding size 32?
It is a balance: big enough to hold patterns, small enough to avoid overfitting.

Q6. What is implicit feedback?
Behavior like purchases, with no explicit ratings.

Q7. What are positive samples?
Customer and product pairs that were purchased, with label 1.

Q8. What is negative sampling?
Random products the customer did not buy, with label 0 (4 per customer here).

Q9. Why is negative sampling needed?
With only positives, the model would learn to predict 1 for everything.

Q10. Why keep only completed orders with quantity > 0?
Cancelled or returned orders are not real purchases.

Q11. Why dropna on the columns?
Missing values break the id mapping and the model.

Q12. What does pd.factorize do?
It maps ids to integers 0 to n-1, which the Embedding layer needs.

Q13. What does the Concatenate layer do?
It joins the user and product vectors into one vector.

Q14. Why a sigmoid output?
It gives the probability that the customer buys the product.

Q15. Which loss is used?
Binary crossentropy.

Q16. What is AUC?
The chance that a random positive scores higher than a random negative. 0.5 is random, 1 is perfect.

Q17. Why AUC and not accuracy?
Negatives are 4 times the positives, so accuracy would be misleading.

Q18. How are recommendations made?
Score all products the customer has not bought, sort high to low, take the top 5.

Q19. What is the cold start problem?
New users and new products have no embedding yet, so the model cannot recommend well.

Q20. Matrix factorization vs this model?
Matrix factorization uses a dot product. This model uses Dense layers, so it learns non-linear interactions.

Experiment 11: Image Captioning (InceptionV3 + LSTM)

Q1. What is image captioning?
Generating a sentence that describes an image.

Q2. What is Flickr8k?
8,091 images with 5 captions each. Split: 6,000 train, 1,000 dev, 1,000 test.

Q3. What is the model structure?
A CNN encoder (InceptionV3) and an LSTM decoder.

Q4. Why InceptionV3?
It is pretrained on ImageNet, so its features are strong. This is transfer learning.

Q5. Why 2048 features?
That is the size of the last global pooling layer of InceptionV3.

Q6. What does pooling='avg' do?
It applies global average pooling and gives one 2048 vector per image.

Q7. Why 299 x 299 and preprocess_input?
InceptionV3 expects that size and scales pixels to -1 to 1.

Q8. Why extract features once?
The CNN is not trained here. Saving the features saves a lot of time.

Q9. Why startseq and endseq?
They mark where a caption begins and ends.

Q10. How are captions cleaned?
Lowercase and remove punctuation.

Q11. Why fit the Tokenizer on training captions only?
To avoid leaking validation or test information.

Q12. What is the vocabulary size V?
Number of words + 1 (index 0 is for padding).

Q13. What is max length L?
The longest caption length. All sequences are padded to it.

Q14. How are training pairs made?
Image + the first i words is the input and word i+1 is the target. A caption of n words gives n-1 pairs.

Q15. Why use add() to merge?
It combines the image vector and the text vector into one vector of equal size.

Q16. What does mask_zero=True do?
It makes the LSTM ignore the padded positions.

Q17. Why Dropout 0.3?
It reduces overfitting.

Q18. How is a caption generated?
Start with startseq, predict a word, append it, repeat until endseq or max length.

Q19. Greedy vs beam search?
Beam search keeps several candidates and usually gives better captions.

Q20. How to evaluate and improve?
Use the BLEU score. Improve with attention or Transformers.